MILITARY SPECIFICATION

# VALVE, LUBRICATING OIL, BACK PRESSURE REGULATING, QUIET-DESIGN, FOR SUBMARINE SYSTEMS

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers quiet-design, self-contained, back pressure regulating valves.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

MILITARY
MIL-V-3 - Valves, Fittings; and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-L-17331 - Lubricating Oil, Steam Turbine (Non-Corrosive).

STANDARDS

MILITARY
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, Center Building, SEC 6124, Prince George's Center, Hyattsville, Maryland 20782 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B1.12-72 - Class 5 Interference - Fit Thread.
B16.5-73 - Steel Pipe Flanges, Flanged Valves, and Fittings.
B18.2.1-72 - Square and Hex Bolts and Screws, Including Askew Head Bolts, Hex Cap Screws, and Lag Screws.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A105-75 - Forgings, Carbon Steel for Piping Components.
A193-75 - Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service.
A194-74 - Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service.
A216-75 - Carbon-Steel Castings Suitable for Fusion Welding for High-Temperature Service.
A217-75 - Martensitic Stainless Steel and Alloy Steel Castings for Pressure-Containing Parts Suitable for High-Temperature Service.
A230-71 - Steel Wire, Oil-Tempered Carbon Valve Spring Quality.
A231-68 - Chromium-Vanadium Alloy Steel Spring Wire.
A232-74 - Chromium-Vanadium Alloy Steel Valve Spring Quality Wire.
A276-75 - Stainless and Heat-Resisting Steel Bars and Shapes.
A313-70 - Chromium-Nickel Stainless and Heat-Resisting Steel Spring Wire.
A351-75 - Ferritic and Austenitic Steel Castings for High-Temperature Service.
B164-70 - Nickel-Copper Alloy Rod and Bar.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

## 3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production a sample shall be examined and tested as specified in 4.3 (see 6.3).

3.2 Valve description. Valves shall be direct spring loaded with oil pressure from an external sensing connection opposing spring compression and directly positioning a single-seated valve element. The sensing element may be a diaphragm, piston, or valving element, providing the design meets the performance requirements of 3.6. The valves shall have a closed spring housing, with a leakoff connection to prevent pressure buildup. A handwheel operated manual override shall be provided to open the valve.

3.3 Materials. Materials shall be as specified in table I. Materials shall be selected to prevent galling, seizing, or excessive wear between parts.

TABLE I. List of materials.

| Name of part | Materials |
|---|---|
| Body, bonnet, and yoke | ASTM A105-75, grade 11<br>ASTM A216-75, grade WCB<br>ASTM A217-75, grade WC1, WC6<br>ASTM A351-75, grade CF3, CF3M, CF8, CF8M |
| Disc and seat ring | ASTM A276-75, types 302, 303, 304, 316, 410, 431, 440<br>ASTM A351-75, grade CF3, CF3M, CF8, CF8M<br>ASTM B164-70<br>QQ-N-288 |
| Stem | Hardened corrosion-resisting steel (400 series) |
| Springs | ASTM A231-68, A232-74, A230-71, A313-70 |
| Body bolts and nuts | ASTM A193-75, grade B7 or B16<br>ASTM A194-74, grade 2H or 4 |



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Note: For all production parts of material whose mechanical properties can be altered by heat treatment, sample material (of the same form as used to fabricate the production parts being heat treated and representing each heat of material used) shall be subjected to tensile tests in the final heat-treated condition for the part. As a minimum, the tensile tests shall determine ultimate strength, yield strength, percent reduction in area, and percent elongation. Tensile test samples shall have been subjected to the same heat treatment as the part(s) made from the material. In the case of parts which contribute to the structural integrity of the pressure boundary, each production heat treatment furnace charge shall include sample material of the same form as used to fabricate the production parts being heat-treated and representing each heat of material in the charge. For other parts, it is not necessary that sample material accompany each furnace charge; however, furnace temperature controls for heat treatment of parts shall be as accurate as the controls of the furnace(s) in which the sample material was heat-treated. Acceptance criteria for the mechanical properties shall be in accordance with the material specification. The tensile test requirements and acceptance criteria for mechanical properties shall be included on the part drawing. In no case shall ductility less than 10 percent elongation in 2 inches be allowed.

3.4 Design and construction.

3.4.1 Design concept. Valves will be operated, maintained, and repaired onboard Naval ships and shall emphasize simplicity, maintainability, ruggedness, and reliability. Design shall permit access for adjustment and repair when working from either side of the valve and without requiring removal of the valve body from the line.

3.4.2 Maintainability. Internal parts shall permit easy disassembly and reassembly with standard tools and shall prevent, as far as practical, the incorrect reassembly of parts. Positioning and alinement of all parts in assembly shall employ positive means so that correct reassembly is repeatedly assured. In no case shall parts for a given valve be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.4.2.1 Valve design shall permit accomplishment of the following maintenance actions within the time limits specified:

| Action | Time allowed |
|---|---|
| Renew main valve trim | 3/4 hour |
| Remove, clean, and replace quiet flow regulating element | 2-1/2 hours |

3.4.3 Interchangeability. Valve design shall permit interchangeability without individual modification of like parts between all valves. Each part shall have part number identity and shall be replaceable from stock or the manufacturer on a non-selective and random basis. Parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation (physical) and performance (function). Physically interchangeable assemblies, components, and parts are those which are capable of being readily installed, removed, or replaced without alteration, misalinement, or damage to parts being installed or to adjoining parts. No fabrication operations such as cutting, filing, drilling, reaming, hammering, bending, prying, or forcing shall be required. This is not intended to preclude the use of special tools, fixtures, and other shop aids during original assembly of the parts into the article. However, any special tools required for maintenance shall be supplied with the valve. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.4.4 Springs. Springs shall not be fully compressed during any normal operation or adjustment of the valve. The working stress shall be such that relaxation shall not exceed 5 percent over a 1000 hour period at the nominal operating temperature. Spring ends shall be squared and ground.

3.4.5 Threads. Threads shall conform to Handbook H28. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. Pipe threads shall not be used. ANSI B18.2.1-72 hex-head standards shall be used for cap screws and bolt heads.



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3.4.6 Pressure-temperature rating. Pressure-temperature rating shall be in accordance with 150 pound series in accordance with ANSI B16.5-73.

3.4.7 End connections. The inlet and outlet end connections shall be in-line 150-pound series, raised-face flanges in accordance with ANSI B16.5-73. End connection size shall be as specified (see 6.2.1). Face-to-face distance between the raised-faces shall be as specified (see 6.2.1). Valves shall be capable of being installed, operated, adjusted, and maintained within the envelope as specified (see 6.2.1). Inlet and outlet bore diameters shall be as specified (see 6.2.1) in order to match the piping bores. The sensing line and leakoff line flanges shall be 1/2-inch iron pipe size (ips), 150-pound series, raised-face flanges in accordance with ANSI B16.5-73.

3.4.8 Spring housing. Spring housing shall be of closed contruction capable of withstanding full system design pressure (see 3.5) without distortion or damage. Spring housing shall be flanged for attachment to the body. The joint shall be secured by either of the following:

(a) Through-bolts threaded the entire length and fitted with a nut on each end. Threads on both bolts and nuts shall be class 2 fit in accordance with Handbook H28.
(b) Tap-end studs with interference fit at the tap end and a class 2 fit at the nut end. Interference fit shall be in accordance with ANSI B1.12-72.

Spring housing shall be located by body guiding (i.e., a close tolerance fit between machined diameters on the body and spring housing) rather than depending on studs or bolts for location. Bearing surfaces of nuts and their respective surfaces on the valve shall be finished machined.

3.4.9 Body construction. Valve body shall be machined from a one-piece casting or forging and be of a basic globe configuration with in-line inlet and outlet ports.

3.4.10 Internal trim. The seat ring shall be threaded into the body or held in position by a cage construction. The quiet-flow regulating element shall be confined on both the top and bottom and under sufficient compression to avoid any movement or looseness. A resilient and readily replaceable seat insert shall be incorporated in either the valve plug or the seat ring. Guide support shall insure proper seating of the valve plug and prevent bending or seizing under all operating conditions. Proper alinement and function of all parts shall be maintained with interchangeable parts and under any tolerance stack-up conditions.

3.4.10.1 Quiet-flow regulating element. The element design employed to achieve a low-noise pressure loss shall avoid excessively small flow passages or any other feature jeopardizing operational reliability. The design and method of fabrication and inspection of the quiet-flow regulating element shall assure that the cross-sectional area at the point of greatest restriction of any given flow passage shall not vary by more than plus or minus 3 percent on a unit per unit basis from the nominal point and that the overall flow capacity or the regulating element shall not vary by more than plus or minus 5 percent on a unit per unit basis. It shall be brazed or otherwise fabricated into a one-piece configuration, capable of withstanding installation and operational stresses. Internal passages shall be such that a strainer installed upstream that will block all particles larger than 0.010 inch will protect against clogging. Element cleaning shall not be required more often than annually. Replacement due to blockage, erosion, wear, etc., not more often than once each 5 years. The element shall be capable of being quickly cleaned by backflush or with ultrasonic equipment. Special fixtures necessary for cleaning the element shall be supplied with each valve. The element structure shall be of a solid or continuous nature, not susceptible to media migration.

3.4.11 Valve adjustment. Means shall be provided for adjusting the setting with the valve under pressure. The adjusting screw shall have right hand threads so that clockwise rotation increases set pressure. The adjustment shall be protected from accidental change by a lock nut and cap.

3.5 Service and ambient conditions. The valve shall operate satisfactorily when installed in a system where the service and ambient conditions are as follows:

(a) Fluid - Lubricating oil Navy symbol 2190 TEP of MIL-L-17331.
(b) Range of design pressures and temperatures - As specified (see 6.2.1).
(c) Range of outlet pressures - As specified (see 6.2.1).
(d) Flow demand - Minimum: Tight shutoff as defined in 3.6.3.
Range of sustained flow demands: As specified (see 6.2.1).
(e) Vibration - As defined by type I of MIL-STD-167-1.
(f) Shock - As defined by grade A, class I of MIL-S-901.



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3.6 Performance requirements.

3.6.1 Flow capacity. Valve shall operate smoothly and without chatter at all flow rates within the specified range (see 3.5(d) and 6.2.1) and when set at any point within the required range of adjustment (see 3.6.5 and 6.2.1).

3.6.2 Accumulation. Valve shall be capable of passing rated flow as defined in 3.6.1 at an accumulation not exceeding that specified (see 6.2.1).

3.6.3 Internal leakage. With inlet pressure decreased to the set pressure, leakage shall be checked over a 2-minute period. The outlet flange shall be removed and there shall be no visible leakage.

3.6.4 External leakage. There shall be no visible external leakage when the valve is subject to an inlet pressure up to and including the proof pressure.

3.6.5 Set pressure adjustment range. Valve shall be adjustable over the set pressure range as specified (see 6.2.1).

3.6.6 Noise. Valve shall meet the airborne and structureborne noise requirements of MIL-STD-740. The contractor shall obtain airborne and structureborne noise measurements in accordance with the requirements, measurements, and data reporting procedures as specified in MIL-STD-740 and the Appendix to this specification. This equipment shall not exceed the grade C airborne and type 2 structureborne criteria of MIL-STD-740.

3.6.7 Mechanical shock and vibration. Valve shall meet the mechanical shock requirements defined by grade A, class I of MIL-S-901 and the environmental vibration requirements defined by type I of MIL-STD-167-1.

3.7 Marking.

3.7.1 Body markings. The manufacturer's name or trademark and the body material composition shall be cast or forged integral with the valve body. The size, rating, and a flow arrow shall be cast or forged integral with the valve body or die stamped on raised metal pads (1/8 inch added wall thickness minimum).

3.7.2 Identification plates. Each valve shall have an identification plate permanently attached to an exposed position on the valve. The identification plate shall be made of corrosion-resistant material and shall contain the following information or a space therefor:

(a) Manufacturer's name and address.
(b) MIL-V-24544.
(c) ANSI 150-pound rated.
(d) Adjustable set pressure range.
(e) Manufacturer's model and part number or identification.
(f) Manufacturer's drawing numbers (assembly and CD).
(g) Technical manual number.
(h) 9-digit CID/APL number.

3.8 Technical data. The contractor shall prepare engineering drawings and a technical manual in accordance with the data ordering documents included in the contract or order (see 6.2.2) and as specified in 3.8.1 through 3.8.3.

3.8.1 Drawings. In addition to the drawing content required by the data ordering document (see 6.2.2), the following unique technical features shall be included:

(a) Calculated average fluid velocity through various portions of the valve. These shall include the inlet and outlet ports and, at minimum, enough sections within the valve to cover any velocity change in excess of 25 percent.
(b) Construction details of the "quiet" throttling element, including: (1) data on maximum particulate sizes that can be passed without clogging, (2) brief discussion or rationale on the mechanics by which pressure breakdown is achieved without generating the noise level normally associated with orificial throttling, (3) method of fabrication: configuration and process of fabrication shall be described in sufficient detail to illustrate that adequate structural strength and freedom from flow passage obstruction is assured during production.
(c) Calculations verifying compliance with the flow passage and overall flow-rate consistency requirements specified in 3.4.10.1 Required dimensional inspections shall be detailed.



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(d) Complete production details of the seat, disc, and stem assembly and all other replaceable internal trim.
(e) Layout of the pressure-containing envelope (body, bonnet, and bottom cap) giving dimensions which control compression of the spiral-wound gaskets. This is to assure that where remachining is necessary to repair the gasket-sealing surfaces on these parts, that compensating cuts can be accurately made to restore original gasket compression. This layout shall also specify the dimensional limits of such corrective remachining within which function of the valve remains unaffected.
(f) Recommended assembly torques, or equivalent procedures, for making up all joints and threaded assemblies.
(g) Tabulation of required gasket characteristics including all dimensions (with tolerances) and load versus compression characteristics (with tolerances).
(h) Mark areas to be radiographic, magnetic particle, or dye penetrant inspected.

3.8.2 Limited rights legend. When the Government has only limited rights in the data shown on the drawings, as determined by the contractual provisions regarding rights in technical data, the drawings furnished may be marked with the following restrictive legend:

"Furnished under United States Government Contract No._____. Shall not be either released outside the Government, or used, duplicated, or disclosed in whole or in part for manufacture or procurement, without the written permission of _____, except for: (a) emergency repairs or overhaul work by or for the Government, where the item or process concerned is not otherwise reasonably available to enable timely performance of the work; or (b) release to a foreign government, as the interests of the United States may require; provided that in either case the release, use, duplication or disclosure hereof shall be subject to the foregoing limitations. This legend shall be marked on any reproduction hereof in whole or in part."

3.8.3 Technical manual. In addition to the general requirements covered by the data ordering document (see 6.2.2), the following unique technical feature shall be included:

(a) Table listing wrench sizes and assembly torques (or other equivalent procedures) for making up all joints and threaded assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.6).

4.3 First article inspection. The first valve furnished under a contract or order shall be subjected to the examinations and tests specified in 4.4 through 4.5.8. Unless otherwise specified herein, the test fluid shall be symbol 2190 TEP lubricating oil of MIL-L-17331. If the valve fails to meet the performance requirements, exhibits leakage, shows evidence of excessive wear or degradation, fails to meet interchangeability requirements, or fails to meet the noise requirements, the valve shall have failed first article examination and testing.

4.4 Examination prior to testing. The first article valve shall be disassembled and visually and dimensionally examined to determine conformance with the requirements of this specification and the valve drawings. Particular emphasis shall be placed on the dimensions, finishes, and condition of the guiding and seating surfaces. Any difficulty encountered in assembly or disassembly shall be noted.



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4.4.1 The maintenance actions specified in 3.4.2.1 shall be demonstrated.

4.5 Tests.

4.5.1 Nondestructive tests. Nondestructive tests shall be as specified in MIL-STD-278 and in accordance with MIL-STD-271. Acceptance criteria shall be in accordance with MIL-STD-278.

4.5.2 Hydrostatic test. Valve shall be pressurized with fresh water or 2190 TEP lubricating oil to the cold test pressure specified in ANSI B16.5-73 for 150 pound series. There shall be no external leakage, permanent distortion, or structural failure. The sensing element may be shunted during this test to prevent damage.

4.5.3 External leakage test. Design pressure (see 3.5) water or 2190 TEP lubricating oil shall be applied to the valve inlet for 2 minutes. There shall be no external leakage, with the exception of the leakoff connection.

4.5.4 Functional test. Valve shall be subjected to the tests specified in 4.5.4.1 through 4.5.4.3. Upstream pressure variation shall at no time exceed that permitted in 3.6.2. There shall be no evidence of hunting, chattering, or any other unstable or unsatisfactory operation of the valve either continuous or transient, at any time during these tests. Valve shall be monitored for noise measurements during these tests, and at all times shall meet the noise criteria of 3.6.6.

4.5.4.1 Accumulation test. Valve shall be set at the mid-point of the specified set pressure range (see 3.6.5 and 6.2.1). Verification shall be made by reducing inlet pressure from a flow condition and checking for leakage. Inlet pressure shall be increased by the amount specified in 3.6.2. Flow rate shall be not less than the maximum specified requirement (see 3.5 and 6.2.1).

4.5.4.2 Seat tightness test. Set pressure shall be established by increasing inlet pressure until the valve passes the maximum specified requirement (see 3.5 and 6.2.1). Set pressure for the purposes of this seat tightness test is defined as an amount below this full-flow pressure equal to the accumulation allowed in 3.6.2. Inlet pressure shall be reduced to the set pressure and the outlet flange shall be removed to measure leakage. There shall be no visible leakage over a 2-minute period.

4.5.4.3 Endurance test. Valve shall be set at the mid-point of the specified set pressure range (see 3.5 and 6.2.1). Flow shall be varied between the rates specified in 3.5 in both increasing and decreasing increments. Valve shall be subjected to a total of 20 hours of operating time by means of this endurance test.

4.5.5 Noise test. Valve shall be airborne and structureborne noise tested to determine conformance to the requirements specified in 3.6.6, under normal operating conditions to duplicate shipboard installation.

4.5.6 High-impact shock test. Valve shall be shock tested to determine conformance to the requirements specified in 3.6.7, and pressurized to the specified design pressure (see 3.5 and 6.2.1) with water or 2190 TEP lubricating oil.

4.5.7 Vibration test. Valve shall be vibration tested to determine conformance to the requirements specified in 3.6.7, and pressurized to the specified design pressure (see 3.5 and 6.2.1) with water or 2190 TEP lubricating oil. There shall be no resonance in the range of frequency tested.

4.5.8 Post test examination. After completion of the tests specified in 4.5.1 through 4.5.7, the test valve shall be disassembled and visually and dimensionally examined. Any damage, excessive wear, or signs of galling or pitting shall be cause for rejection.

4.6 Quality conformance inspection. Each production valve shall be examined and tested as specified in 4.6.1 through 4.6.5.

4.6.1 Visual and dimensional examination. Each valve shall be visually and dimensionally examined to verify conformance with the requirements of this specification not requiring tests and the engineering drawings.

4.6.1.1 The quiet-flow regulating element shall have undergone, during production, the inspections detailed in 3.8.1(c). A complete record of the inspection results for each element shall be supplied with the valve (or element).



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4.6.2 Nondestructive tests. Nondestructive tests as specified in 4.5.1 shall be conducted on each valve.

4.6.3 Hydrostatic test. Hydrostatic test as specified in 4.5.2 shall be conducted on each valve.

4.6.4 Seat tightness test. Seat tightness tests as specified in 4.5.4.2 shall be conducted on each valve.

4.6.5 External leakage test. External leakage test as specified in 4.5.3 shall be conducted on each valve.

4.7 Inspection of preparation for delivery. The preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

## 5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation-packaging, packing, and marking. Valves shall be individually preserved-packaged level A or C, packed level A, B, or C and marked in accordance with MIL-V-3, as specified (see 6.2.1).

5.1.1 Special marking. In addition to the marking, as applicable, required in 5.1 and 5.1.2, shelf life marking shall be applied. Shipping containers (see 5.1) shall be stenciled in accordance with the marking instructions of MIL-STD-740.

5.1.2 Repair parts. Repair parts shall be preserved-packaged, packed, and marked (see 5.1.1) in accordance with the levels of MIL-R-196 as applicable for the intended use and destination as follows:

| | Levels | |
|---|---|---|
| | Preservation-packaging | Packing |
| Onboard (accompanying valves) | A | C |
| Stock | A | B |
| Immediate use | C | C |

Unless otherwise specified (see 6.2.1), repair parts shall be packed separately and shipped concurrently with the valve(s).

5.2 Use of polystyrene (loose-fill) material.

5.2.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard, or transparent flexible cellular material."

5.2.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler, and dunnage.

5.3 Unpacking instructions. Unpacking instructions, as specified in MIL-STD-740, shall be provided.



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5.4 Technical manuals. Technical manuals, which accompany shipments that are packed level A or B, shall be packaged in a transparent waterproof plastic bag, minimum 4 mil thick. Closure shall be by heat sealing. Technical manuals shall not be placed within any flexible sealed barrier enclosing components. The copy(s) of the manual shall be placed in the shipping container housing the main unit. Packing lists shall indicate which container contains the technical manual(s) and shall also state the approximate location therein. For ease of removability the location of the manual shall be such that it is readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specification or packed in containers conforming to the requirements for level A, B, or C, as specified (see 6.2.1).

5.5 Data (see 3.8). Data shall be prepared for delivery in such a manner as to insure the required information is protected against deterioration, physical damage or loss during shipment from the contractor to the receiving activity. Packages or shipping containers shall as a minimum conform to the level C requirements of 5.1.

6. NOTES

6.1 Intended use. The quiet-design lubricating oil back pressure regulating valve specified herein is intended for service in a submarine lubricating oil system.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) End connection size (see 3.4.7).
(c) Face-to-face distance (see 3.4.7).
(d) Maintained envelope (see 3.4.7).
(e) Bore diameters (see 3.4.7).
(f) Range of design pressures and temperatures (see 3.5, 4.5.6, and 4.5.7).
(g) Range of outlet pressures (see 3.5).
(h) Range of sustained flow demands (see 3.5, 4.5.4.1 and 4.5.4.2).
(i) Accumulation (see 3.6.2).
(j) Adjustable set pressure range (see 3.6.5, 4.5.4.1, and 4.5.4.3).
(k) Special features required.
(l) Applicable levels of preservation-packaging and packing required (see 5.1).
(m) Levels of preservation-packaging and packing required for repair parts (see 5.1.2).
(n) Use of polystyrene (loose-fill) material (see 5.2.1).
(o) Packing of technical manuals (see 5.4).

6.2.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

| Paragraph | Data requirements | Applicable DID | Option |
|---|---|---|---|
| 3.8.1 | Drawings, engineering and associated lists, Level 1, (conceptual and developmental design) | DI-E-7013 | ------ |
| 3.8.3 and 5.4 | Technical manuals | DI-M-2043 | Type I of MIL-M-15071 |

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Definitions. The definitions specified in 6.4.1 through 6.4.6 shall apply to the various technical terms wherever such terms appear in this specification.

6.4.1 Working pressure and temperature. The pressure and temperature which the valve is subjected to under normal working conditions.

6.4.2 Design pressure and temperature. The maximum pressure and temperature conditions which the valve is subjected to in service, and upon which the strength of the pressure containing envelope is based.

6.4.3 Proof pressure. The maximum test pressure that the valve should withstand without damage. Valve operation is not required during application of the proof pressure, but following proof test the valve should be capable of meeting all specification requirements.

6.4.4 Set pressure. The pressure at which the valve seats tightly (defined as a condition where flow is reduced to a point where it is equal to or less than the allowable leakage) (see 3.6.3) on decreasing inlet pressure. Expressed in $lb/in^2$ gage pressure.

6.4.5 Set pressure range. The span of set pressures over which the valve can be adjusted while meeting all performance requirements and without changing any parts.

6.4.6 Accumulation. The increase in inlet pressure, above the set pressure, required for the valve to pass rated flow. Expressed in $lb/in^2$.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH

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# APPENDIX

## NOISE TESTING DETAILS

10. SCOPE

10.1 Scope. This appendix details noise testing for quiet design valves.

20. APPLICABLE DOCUMENTS

20.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this appendix to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-17185 - Mounts, Resilient, General Specifications and Tests for (Shipboard Application).
MIL-M-17508 - Mounts, Resilient: Types 6E2000, 6E900, 6E900BB, 7E450, 7E450BB, 6E150, and 6E100.
MIL-M-19379 - Mounts, Resilient, Mare Island Types 11M15, 11M25, and 10M50.
MIL-M-19863 - Mounts, Resilient: Type 5B5000H.
MIL-M-21649 - Mount, Resilient, Type 5M10,000-H.

DRAWING

MILITARY

NAVSHIPS 810-2145600 = Mounts, Resilient, Unbonded, Spool, Portsmouth Type.

PUBLICATION

MILITARY

NAVSHIPS 0939-LP-001-2010 - A Guide to Structural Damping for Naval Applications.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. QUIET VALVE TEST FACILITY

30.1 The test facility shall be capable of reproducing the shipboard operational conditions for the valve being testing. Operational parameters such as pressure, temperature, flow rate, flow velocity, etc., shall be as specified in this specification. The valve shall be connected between straight pipe of appropriate schedule and nominal diameter for which the valve pipe connections were designed. In order to minimize system noise influence on the valve and to provide a standardized basis for comparison, the straight pipe sections shall be at least 10 pipe diameters in length and shall be damped in accordance with Navy pipe damping procedures of NAVSHIPS 0939-LP-001-2010 for at least 10 pipe diameters. In order to reduce system noise, and when conditions permit their use, flexible hoses, acoustic filters, or other noise attenuating or isolating devices may be incorporated in the piping system at distances greater than 10 pipe diameters from the valve. System background noise should be at least 10 dB below the valve's acceptance criteria. Piping needing support within 10 pipe diameters of the valve being tested shall be isolation mounted from its supports. Generally, low frequency mounts will be used in accordance with MIL-M-17508, MIL-M-19379, MIL-M-19863, MIL-M-21649, and Drawing 810-2145600. Where other than Navy standard mounts are used, they shall be in accordance with MIL-M-17185. However, distributed isolation material, such as Isomode, or equal, if considered sufficient to eliminate influence of the test facility from background noise sources, may be used.

40. QUIET VALVE TRANSDUCER LOCATIONS

40.1 Transducer locations shall be selected so as to determine as closely as possible the vibratory excitation that would be transmitted by the valve to the ship either through the piping system or foundation. For those valves which are supported only by piping, the transducers shall be located on the appropriate inlet(s) and outlet(s) connections of the valve. Structureborne noise measurements shall be made at these locations in directions



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perpendicular and parallel to the flow at the piping connections. For valves which are also supported by foundations, additional transducer locations at the attachment points (feet) of the valve are required. Transducers shall be located on the base of the valve at two diagonally opposite corners above mounts and as close as possible to the valve's designed attachment points. At least two mutually perpendicular measurement directions are required for each location.

50. QUIET VALVES FOR TEST

50.1 In all cases, the quiet valve shall be oriented in its normal operating position and supported by its designed piping attachment points directly to the hard piping without flexible pipe connections. Foundations and supports shall be provided for those valves which, because of the size and weight, require same in shipboard systems. These valves shall be isolation mounted to their foundations during the noise test.



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No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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